TROUBLESHOOTING TOOLS

Networking Assignment

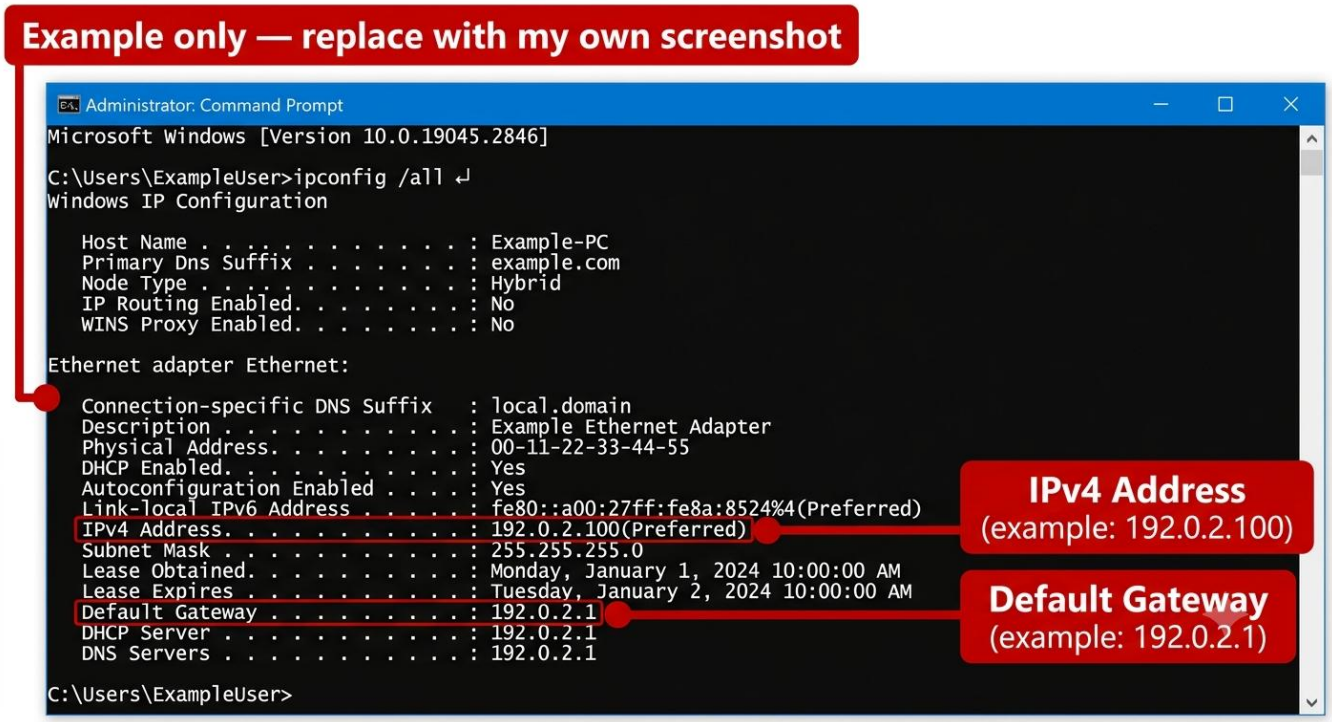
Task 1: Find the IPv4 Address and Default Gateway

Command: ipconfig

Steps:

1. Open the Start menu and search for Command Prompt (CMD).
2. Open Command Prompt and type ipconfig, then press Enter.
3. Find the active Wi-Fi or Ethernet adapter. Note its IPv4 Address and Default Gateway.

Result to fill in after running the command:



Task 2: Check Connectivity to Google

Command: ping www.google.com

Steps: In Command Prompt, type the command above and press Enter. Windows sends packets to Google and displays replies, packet loss, and approximate response times. At the bottom, look for “Average” in the round-trip time summary.

Result to fill in:

Note: The response time depends on your connection and may differ each time. If the command fails, it does not always mean the Internet is completely unavailable; some networks block ping replies.

Task 3: Trace the Route to Flipkart

Command: tracert www.flipkart.com

Steps: Run the command in Command Prompt and wait for the trace to finish. Each numbered line is a hop (a router or other network point that responds along the route). Count the numbered hop lines. If some lines show asterisks, that device may not respond to traceroute probes.

Task 4: Find Spotify's IP Address with nslookup

Command: nslookup www.spotify.com

Steps: Run the command in Command Prompt. Read the Address field or fields shown in the answer section. A website can return more than one IP address, and the results may vary because of DNS, caching, or location.

What does nslookup verify? It checks whether DNS can resolve a domain name, such as www.spotify.com, into an IP address.

Task 5: Troubleshoot a Laptop That Cannot Open Zomato

I would run these three commands in order:

Command	What I would check	What the result may tell me
1. ipconfig	Check the laptop's IPv4 address, subnet details, and default gateway.	If the address or gateway is missing or incorrect, the laptop may not be connected properly to the local network.
2. nslookup www.zomato.com	Check whether DNS resolves Zomato's domain to an IP address.	If name resolution fails, the problem may involve DNS settings or the DNS service.
3. ping www.zomato.com	Check whether the destination responds over the network.	Replies suggest reachability; no replies may indicate a network issue, server filtering, or blocked ICMP. Ping alone cannot prove the website is down.

Conclusion

The ipconfig command checks local IP configuration, ping tests basic network reachability and response time, tracert shows the route taken through network hops, and nslookup checks DNS name resolution. Using these commands in a logical order helps narrow down whether a problem is related to the local network, DNS, or the route to a website.